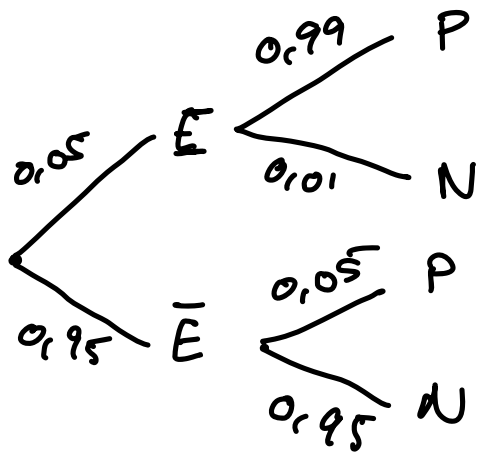
1.



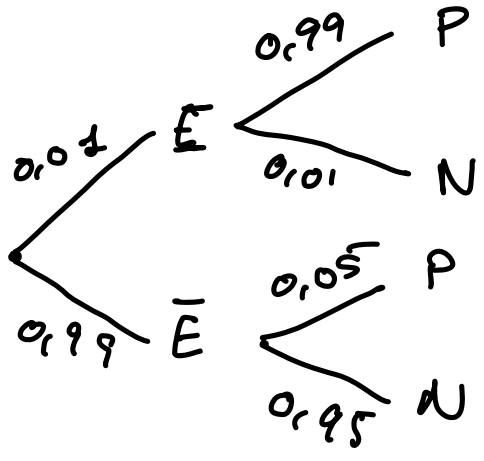
a) $P(E|P) = \frac{P(E \cap P)}{P(P)} = \frac{0,05 \cdot 0,99}{0,05 \cdot 0,99 + 0,95 \cdot 0,05}$

$P(E|P) = 0,51$ $\quad \searrow 0,097$

b) $P(\bar{E}|\bar{P}) = \frac{0,95 \cdot 0,95}{1 - 0,097} = 0,999\ldots$

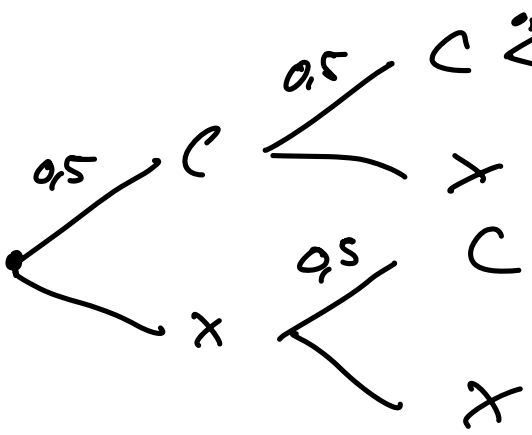
c) $P(\text{correcto}) = 0,05 \cdot 0,99 + 0,95^2 = 0,952$

d)



$P(E|P) = 1/6$

2.



0,5 Par
Impar Ganar = 2 caras + Num Par
y 1 cara + 5 ómos

a) $P(\text{Ganar}) = 0{,}5^3 + (0{,}5^2 \cdot 2/6 \cdot 2) = \frac{7}{24} = 0{,}292$

b) $P(C \cap C \mid G) = \frac{P((C \cap C) \cap G)}{P(G)} = \frac{0{,}5^3}{7/24} = 3/7$

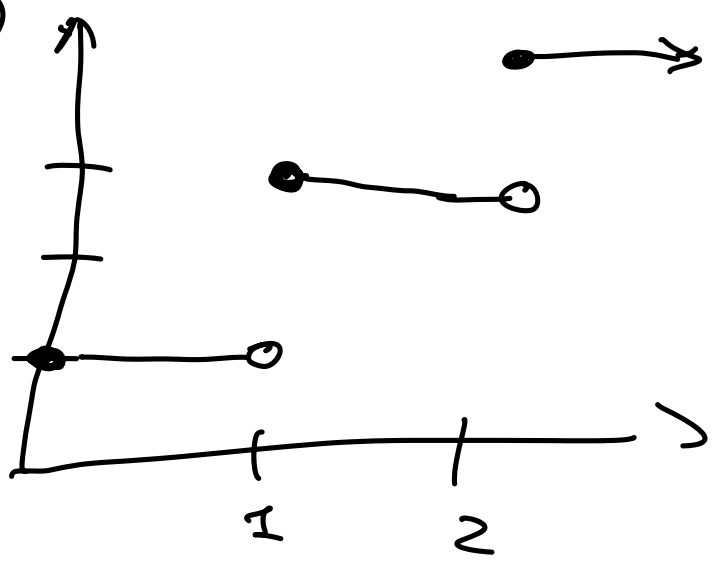
c) $P(5 \mid G) = \frac{P(5 \cap G)}{P(G)} = \frac{0{,}5^2 \cdot 1/6 \cdot 2}{7/24} = 2/7$

3.

a)

x_i	$P(X = x_i)$
0	0,25
1	0,5
2	0,25

b)



3.1 - $X \sim B(8, 0,5)$

a) $P(X=6) = \binom{8}{6} 0,5^6 0,5^2 = 0,11$

b) $P(X \geq 2) = 1 - P(X \leq 1)$

$P(X \geq 2) = 1 - P(x=0) + P(x=1) = \frac{247}{256}$